Secure wiring and finish the design

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Document permanent connections and evaluate added mass or friction.

SUCCESS CRITERIA - I can...

1. Check circuit continuity and mechanical security.
2. Preserve moving-part clearance during decoration.
3. Relate a design revision to an energy transfer or force.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “strain relief” and “insulation” to explain your model or result. Sentence starters are optional.

strain relief: support that keeps a pull from stressing an electrical joint

insulation: material that limits unwanted electric conduction

short circuit: an unintended low-resistance path bypassing a component

01 Warm-up: bring back an earlier idea

A 40-tooth driver turns a 240-tooth gear.
Give ideal output/input torque and speed ratios and state the loss assumption.

Word help: torque = the turning effect of a force about an axis

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: ① ② ③

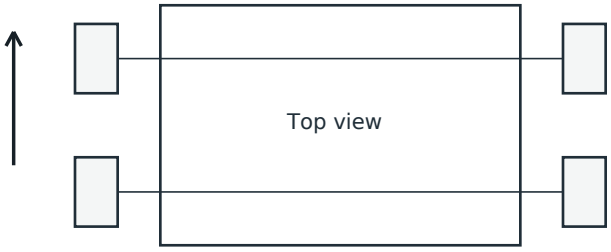
02 Worked example: explain the model

Reference: I = 0 with power disconnected. Disconnect before soldering/securing wires. Use insulation and strain relief; route wires clear of gears. Follow supervised tool procedures.

Worked reasoning: A secure circuit also needs strain relief: a pull on a wire should not be carried solely by a solder joint. Insulation prevents exposed conductors from making unintended contacts.

Your turn: Draw one wire route showing a connection, insulation, strain relief and clearance from a rotating axle.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Draw your final wiring route.
Mark soldered joints, insulation, strain relief and the switch location.
Identify one possible wheel or gear interference point.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Marzano check - circle: 1 2 3

04 Apply the physics

A simplified model gives the car a constant 0.30 N net forward force.
Compare acceleration at masses 0.20 kg and 0.25 kg.
Complete the table, calculate the percent decrease and state why this fixed-force prediction may differ from a real motor test.

Sentence starter: Both ... . They differ in ... because ... .

Design	Mass (kg)	Net force (N)	Acceleration
Before	____	____	____
After	____	____	____

Marzano check - circle: 1 2 3

05 Explain your reasoning

A wire rubs a wheel.
Explain the possible effect on energy transfer and propose a repair before another powered test.

Sentence starter: The step I would change is ... . The evidence is ... .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Near an ideal long straight wire, current becomes 4 times as large and distance becomes 6 times as large.
Compare B.
Mark the distances and field directions you compare.

Word help: current = rate of charge flow

Sentence starter: Both ... . They differ in ... because ... .

Current out of page (dot). Add field arrows.



Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

Explain why “lighter is always faster” is too strong for a powered car.
Name two other influential factors.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

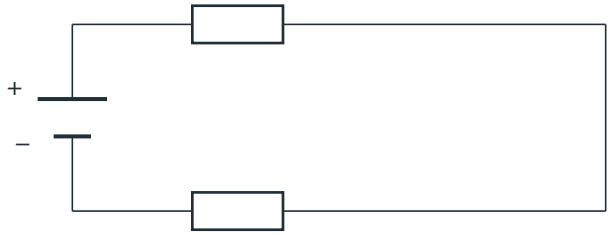


Marzano check - circle: 1 2 3

08 STRETCH: test the model

A wire fault adds 2 ohms in series to a supplied motor model of 4 ohms on 6 V.
Compare current before/after and power in the unwanted resistance.
Explain why a warm joint deserves investigation.

Sentence starter: Both ... . They differ in ... because ... .



Marzano check - circle: 1 2 3